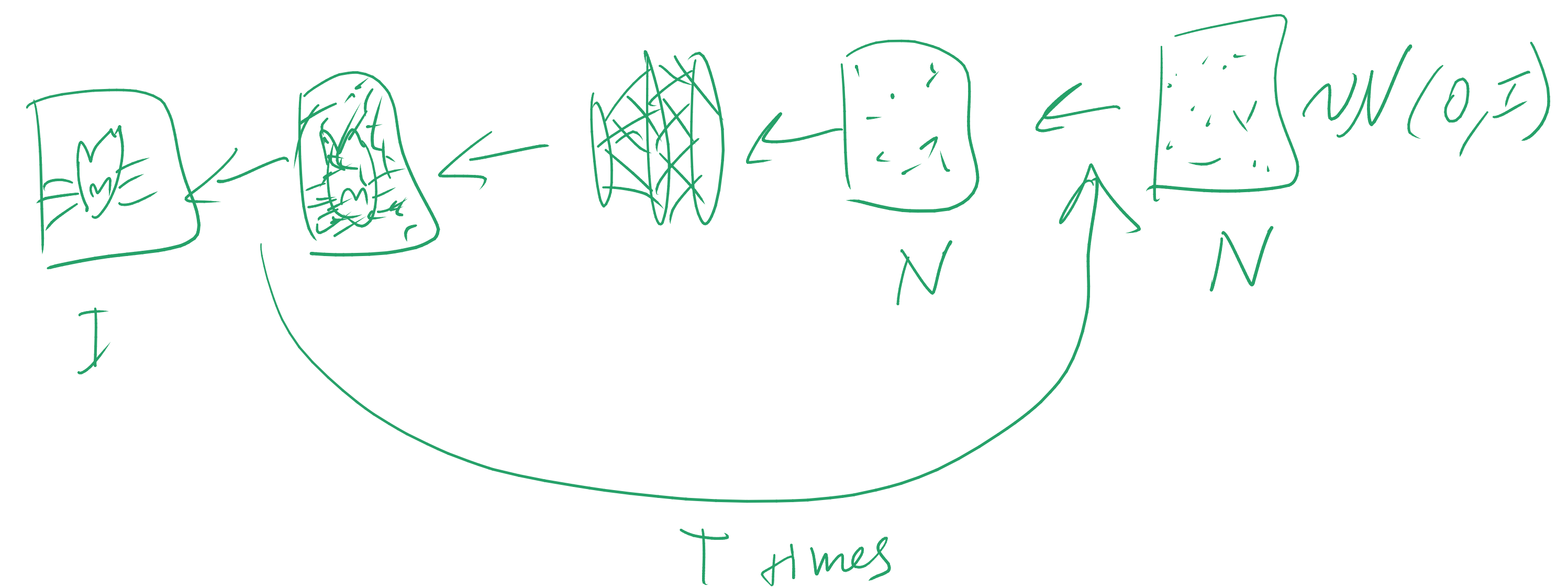
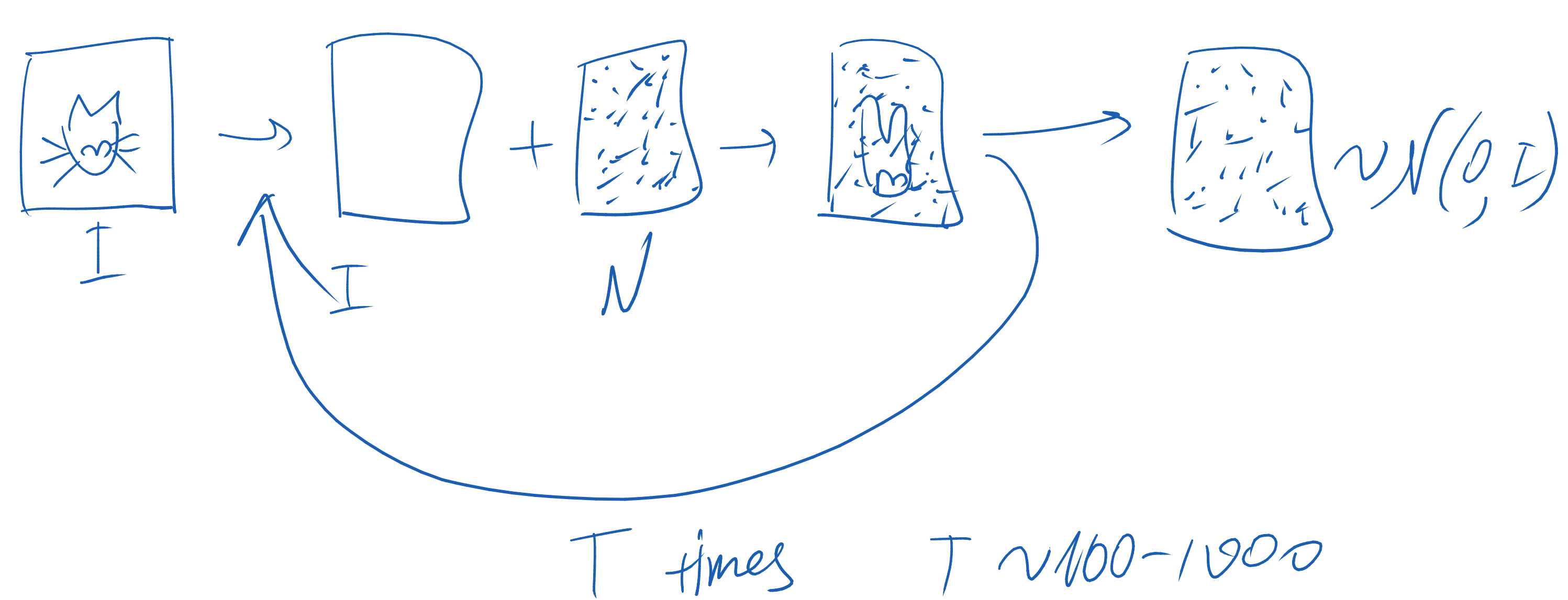




Denoising Diffusion Probabilistic Models (DDPM)

2020

$\mathbb{R}^n$

T - hyperparameter

$0, 1, \dots$

T

$T+1$ step



pdata



$\mathcal{N}(0, I)$

Forward process

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t | \sqrt{1-\beta_t} x_{t-1}, \beta_t I)$$

$$\left(\begin{array}{l} x_{t-1} \sim \mathcal{N}(0, I) \quad x_t = x_{t-1} + \varepsilon \sqrt{\beta_t} \\ \text{Var}(x_t) > \text{Var}(x_{t-1}) \end{array} \right)$$

$$q(x_{1:T} | x_0) := \prod_{t=1}^T q(x_t | x_{t-1})$$

Backward process

$$p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1} | \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

$$p_\theta(x_{1:T}) = p(x_T) \prod_{t=1}^T p_\theta(x_{t-1} | x_t) \quad \text{diag}(\sigma_\theta^2)$$

$\uparrow$
 $\mathcal{N}(0, I)$

$$1) x_T \sim \mathcal{N}(0, I)$$

2) for $t = T+0 \dots 1$:

$$x_{t-1} \sim p_\theta(x_{t-1} | x_t)$$

3) x_0 - final sample

$$x_0 \sim p_{\text{data}}$$

$$\begin{aligned} \log p_\theta(x_0) &= \log \int p_\theta(x_{0:T}) dx_1 \dots dx_T = \\ &= \log \int q(x_{1:T} | x_0) \frac{p_\theta(x_{0:T})}{q(x_{1:T} | x_0)} dx_1 \dots dx_T = \end{aligned}$$

$$= \log \mathbb{E}_{q(x_{1:T} | x_0)} \frac{p_\theta(x_{0:T})}{q(x_{1:T} | x_0)} \geq \left. \begin{array}{l} \text{Jensen} \\ \text{Inequality} \end{array} \right\}$$

$$\geq \mathbb{E}_{q(x_{1:T} | x_0)} \log \frac{p_\theta(x_{0:T})}{q(x_{1:T} | x_0)} \quad \textcircled{=} \rightarrow \max$$

$$\textcircled{\leq} \mathbb{E}_{q(x_{1:T}|x_0)} \left[\log p(x_T) + \sum_{t=1}^T \log \frac{p_\theta(x_{t-1}|x_t)}{q(x_t|x_{t-1})} \right]$$

$$x_0 \sim p_{\text{data}}$$

||
L

Training (wrong)

1) Sample $x_1 \sim q(x_1|x_0) \dots x_T \sim q(x_T|x_{T-1})$

$$2) \nabla_{\theta} \sum_{t=1}^T \frac{p_\theta(x_{t-1}|x_t)}{q(x_t|x_{t-1})}$$

$$L = \mathbb{E}_{q(x_{1:T}|x_0)} \left[-\text{KL}(q(x_T|x_1) \| p(x_T)) - \sum_{t=2}^T \text{KL}(q(x_{t-1}|x_t, x_0) \| p_\theta(x_{t-1}|x_t)) + \log p_\theta(x_0|x_1) \right]$$

$N(\mu, \Sigma)$

$$q(x_{t+1} | x_t, x_0) = \frac{q(x_0, x_{t+1}, x_t)}{q(x_t, x_0)} =$$

$$= \frac{\cancel{q(x_0)} q(x_{t+1} | x_0) q(x_t | x_{t+1})}{\cancel{q(x_0)} q(x_t | x_0)}$$

$$q(x_t | x_0) = \mathcal{N}\left(x_t \mid \prod_{i=1}^t \sqrt{1 - \beta_i} x_0, \left(1 - \prod_{i=1}^t \sqrt{1 - \beta_i}\right) I\right)$$

$$q(x_{t+1} | x_t, x_0) \sim \mathcal{N}(\mu, \Sigma)$$

Training (final)

x_0

$$1) t \sim U[1, \dots, T], x_t \sim q(x_t | x_0)$$

$$2) \nabla_{\theta} \text{KL}(q(x_{t+1} | x_t, x_0) \| p_{\theta}(x_{t+1} | x_t))$$

(DDIM)

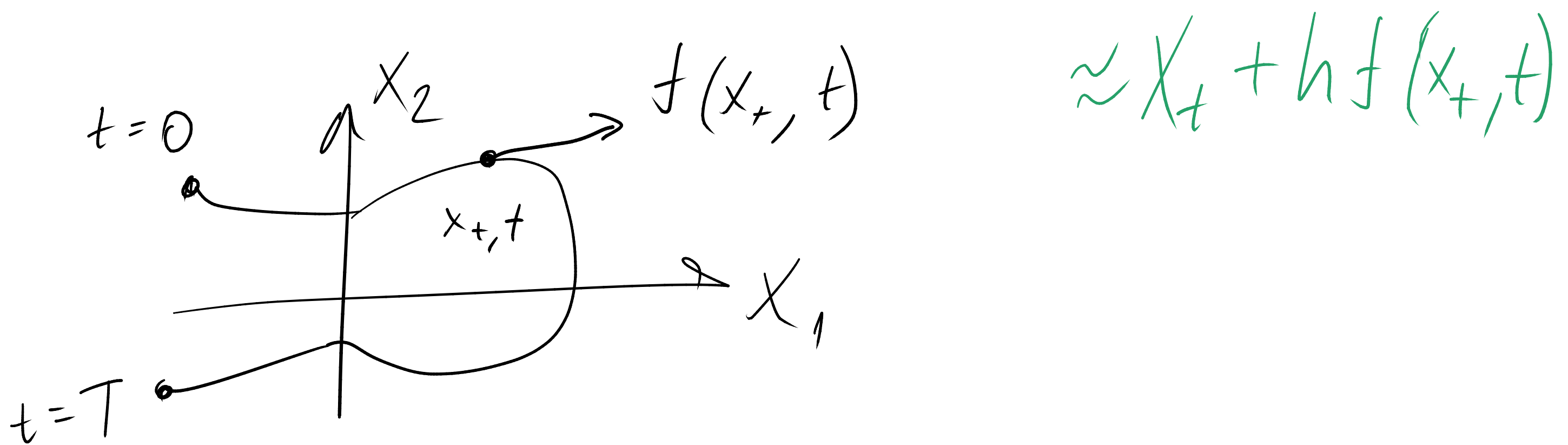
Denoising Diffusion Implicit Models

Ordinary Differential Equation (ODE)

$$X_t \in \mathbb{R}^n \quad t \in [0, T]$$

$$\begin{cases} dx_t = f(x_t, t) dt \\ X_0 = v \in \mathbb{R}^n \end{cases}$$

$$X_{t+h} = X_t + h \cdot f(x_t, t) + \bar{O}(h) \approx$$



Stochastic Differential Equation (SDE)

$$X(t) \quad t \in [0, T]$$

$$\begin{cases} dX_t = f(x_t, t) dt + g(t) dW_t \\ X_0 \sim p_0 \end{cases}$$

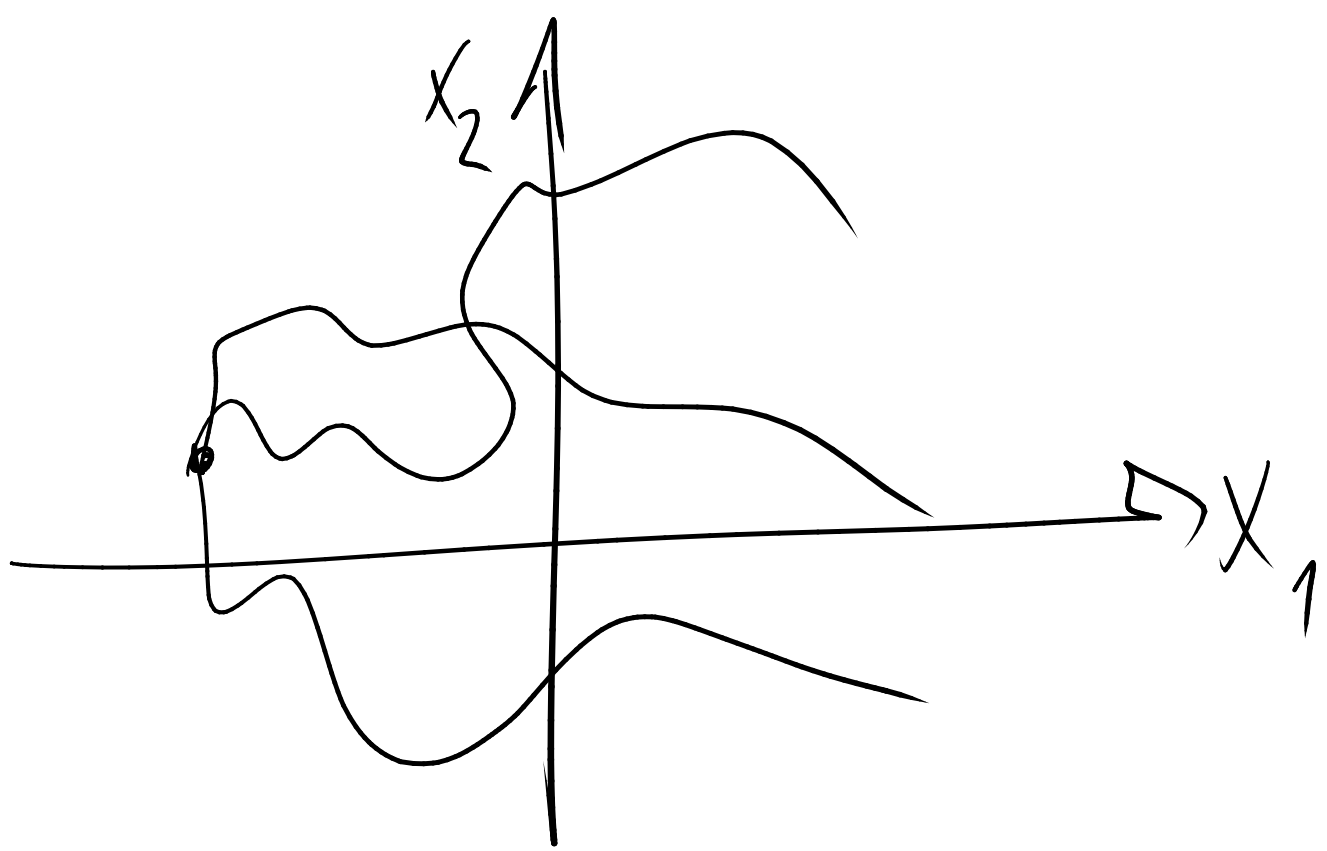
$$dW_t \sim N(\mu, \Sigma)$$

standard
Werner
process

Euler scheme

$$X_{t+h} = X_t + h f(X_t, t) + \sqrt{h} g(t) \varepsilon_t$$

$$\varepsilon_t \sim \mathcal{N}(0, I)$$



$$X_t \sim p_t \leftarrow \text{solution}$$

$$t: 0 \rightarrow T$$

$$\left\{ \begin{array}{l} dX_t = \left[f(X_t, t) - g^2(t) \underbrace{\nabla_x \log p_t(X_t)}_{\text{score function}} \right] dt + g(t) dW_t \\ X_T \sim p_T \\ t: T \rightarrow 0 \end{array} \right.$$

$$X_{t-h} = X_t - h f(t_+, t) + h g^2(t) \nabla_x \log p_+(X_t) + \sqrt{h} g(t) \varepsilon_t$$

Forward SDE:

$$\left\{ \begin{array}{l} dX_t = -\frac{1}{2} \beta(t) X_t dt + \sqrt{\beta(t)} dW_t \\ X_0 \sim p_{\text{data}} \\ X_T \approx N(0, I) \end{array} \right.$$

Reverse SDE:

$$\left\{ \begin{array}{l} dX_t = -\frac{1}{2} \beta(t) X_t dt - \beta_t \nabla_x \log p_t(x_t) dt + \sqrt{\beta(t)} dW_t \\ X_T \sim N(0, I) \end{array} \right.$$

μ_t DDPM $\propto$ score DDIM

h — parameter for number of steps

Denoising Score Matching

$$S_\theta(x, t) \approx \nabla_x \log p_t(x)$$

$$\iint p_t(x) \| s_\theta(x, t) - \nabla_x \log p_t(x) \|^2 dx dt$$

$\rightarrow \min_{\theta}$



$$\mathbb{E} \| s_\theta(x_t, t) - \nabla_x p_t(x_t | x_0) \|^2 \rightarrow \min_{\theta}$$

$$\left\{ \begin{array}{l} t \sim U[0, T] \\ x_0 \sim p_{data} \\ x_t \sim p_t(x | x_0) \end{array} \right.$$